EXPLORATORY DATA ANALYSIS REPORT

Employee Attrition & Workforce Retention Analytics

Project Title	Employee Attrition & Workforce Analytics using Python
Dataset Source	IBM HR Analytics Employee Attrition & Performance (1,470 records)
Core Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn
Student Assignment	Exploratory Data Analysis (EDA)

1. Executive Summary & Problem Definition

Employee turnover (attrition) is a major concern for companies because hiring, onboarding, and training replacement employees requires significant time and financial investment. In addition, when skilled employees depart, teams experience productivity slowdowns and loss of valuable operational knowledge.

The objective of this project is to perform a thorough Exploratory Data Analysis (EDA) on the IBM HR Analytics dataset to uncover why employees leave, which groups are most at risk, and what practical retention steps management can implement. The analysis was conducted in Python using Pandas, NumPy, Matplotlib, and Seaborn.

Key Baseline Finding: Out of 1,470 total employees in the company, 237 employees left, representing an overall voluntary attrition rate of 16.12%. This analysis demonstrates that departures are not random but strongly tied to overtime work, low compensation in junior levels, role dissatisfaction, and long commutes.

2. Data Auditing & Cleaning Decisions

Before beginning visual and statistical analysis, rigorous data auditing and cleaning was performed:

• Missing Values: Checked the entire dataset across all 35 columns. Exactly 0 missing values were found, confirming a complete dataset.

• Duplicates: Checked full row duplicates. Exactly 0 duplicate rows were detected, and EmployeeNumber was confirmed to have 1,470 unique IDs.

• Zero-Variance Columns: Identified three constant columns: 'EmployeeCount' (always 1), 'Over18' (always 'Y'), and 'StandardHours' (always 80). Because these features have zero variance, they provide zero explanatory value and were dropped, reducing the dataset to 32 relevant attributes.

• Domain Integrity: Verified that TotalWorkingYears >= YearsAtCompany across all records (0 errors) and that (Age - TotalWorkingYears) >= 16 across all records (0 errors).

• Target Encoding: Created a numeric binary flag 'Attrition_Flag' (1 = Yes, 0 = No) to allow direct calculation of attrition percentages and correlation statistics.

3. Key Analytical Findings

Through our visualizations and statistical checks, five primary drivers of employee turnover were identified:

1. Overtime is the Leading Operational Driver: Employees working regular overtime suffer an attrition rate of 30.53%, compared to only 10.44% for employees with no overtime. Working overtime increases the risk of quitting by almost 3 times, highlighting workplace fatigue and burnout.

2. Compensation Disparity in Entry Levels: Departing employees earn a median monthly income of $3,202, while staying employees earn a median of $5,204 (a gap of approximately $2,000 every month). Resignations are concentrated in Job Levels 1 and 2, whereas senior leadership roles (Job Levels 4 and 5) experience virtually zero attrition.

3. Frontline Role Vulnerability: Turnover is heavily role-dependent. Sales Representatives show the highest attrition rate in the entire company at 39.76% (nearly 4 out of 10 leave), followed by Laboratory Technicians at 23.94% and Human Resources at 23.08%. In contrast, Research Directors (2.50%) and Managers (4.90%) are very stable.

4. Early-Career Mobility Cliff: Turnover is heavily skewed toward junior staff. Departing staff have a lower median age (32.0 years vs 36.0 years) and much less work experience (median 7.0 years vs 10.0 years). Resignations peak within the first 0 to 3 years at the company.

5. Commute Strain & Work-Life Balance: Employees rating their Work-Life Balance as 1 ('Bad') exhibit an attrition rate of 31.25%, compared to ~14-17% for higher ratings. Similarly, employees who left have a higher average commute distance (10.63 miles vs 8.92 miles), showing that long travel creates extra daily strain.

4. Visualization Quota & Analysis Summary

The project fulfilled and exceeded all required assignment visualization quotas:

Chart Type	Minimum Required	Visualizations Created in Project
Histograms / KDE	3	Age distribution, Total Working Years, Monthly Income distribution
Box Plots	3	Monthly Income by Attrition, Commute Distance by Attrition, Tenure by Job Role
Count Plots	2	OverTime status by Attrition, Work-Life Balance ratings by Attrition
Bar Charts	3	Attrition rate by Department, Attrition rate by Job Role, Attrition by Education Field
Scatter Plots	2	Age vs Monthly Income by Attrition, Total Working Years vs Monthly Income
Correlation Heatmap	1	11 numerical metrics correlation matrix (Tenure, Income, Age, Satisfaction)
Additional Visuals	2	Job Satisfaction rating bar chart, Monthly Income across Job Levels box plot

5. Strategic Recommendations for HR Leadership

Based on the data insights, the following concrete business actions are recommended:

Strategic Initiative	Target Employee Group	Actionable Plan & Expected Outcome

Overtime Capping & Comp-Time	Non-exempt hourly, Sales, & Lab Techs	Audit workload schedules; place strict weekly limits on consecutive overtime; provide paid compensatory time-off. Expected to reduce overtime turnover from 30.5% toward 18-20%.
Entry-Level Pay Benchmarking	Job Levels 1 & 2 (Sales Reps & Lab Techs)	Benchmark entry salaries closer to market median ($3,600-$4,000/mo); introduce milestone loyalty bonuses at 12 and 24 months to address the 0-3 year retention cliff.
Hybrid Commute Flexibility	Employees commuting >10 miles in office roles	Introduce a 2-3 day hybrid remote work policy to eliminate daily travel fatigue and reduce turnover among distant commuters.
Frontline Manager Enablement	Team Leads in Sales & Laboratory teams	Provide training for frontline managers on conducting bi-weekly 1-on-1 check-ins. Tenure with manager is strongly protective against attrition.

6. Conclusion

Conclusion: By addressing the core friction points—overtime fatigue, entry-level compensation, and long commutes—the organization can protect its early-career talent pipeline and significantly reduce avoidable voluntary turnover costs.